

GATE Data Science & Artificial Intelligence (DA)

Comprehensive 80-Question Topic-Wise Question Bank

10 PYQs + 10 Practice Questions with Detailed Step-by-Step Solutions per Topic

Topics: Eigenvalues/Vectors, LDA & Dimensionality Reduction, Timestamping & Deadlocks, Logical Reasoning

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1 Topic 1: Eigenvalues & Eigenvectors

1.1 10 Previous Year Questions (PYQs)

Q1.1

[GATE DA 2024 | MCQ | 1 Mark]

Consider the matrix $M = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$. Which ONE of the following statements is TRUE?

- (A) The eigenvalues of M are non-negative and real.
- (B) The eigenvalues of M are complex conjugate pairs.
- (C) One eigenvalue of M is positive and real, and another is zero.
- (D) One eigenvalue of M is positive and real, and another is negative and real.

Answer: (B) **Solution:** The characteristic equation is $\det(M - \lambda I) = (2 - \lambda)(1 - \lambda) - (-3) = \lambda^2 - 3\lambda + 5 = 0$. The discriminant is $D = (-3)^2 - 4(1)(5) = 9 - 20 = -11 < 0$. Since the discriminant is negative and the matrix entries are real, the eigenvalues form a complex conjugate pair $\lambda = \frac{3 \pm i\sqrt{11}}{2}$.

Q1.2

[GATE DA 2024 | NAT | 2 Marks]

Let A be a 3×3 real matrix with eigenvalues 1, 2, and 3. What is the trace of the matrix $(A^2 - 2A + I)$? **Answer: 6** **Solution:** By the spectral mapping theorem, if λ is an eigenvalue of A , then $f(\lambda) = \lambda^2 - 2\lambda + 1 = (\lambda - 1)^2$ is an eigenvalue of $f(A) = A^2 - 2A + I$. For $\lambda_1 = 1$: $f(1) = (1 - 1)^2 = 0$. For $\lambda_2 = 2$: $f(2) = (2 - 1)^2 = 1$. For $\lambda_3 = 3$: $f(3) = (3 - 1)^2 = 4$. The trace is the sum of the eigenvalues: $\text{trace}(f(A)) = 0 + 1 + 4 = 5$? Wait: $(1 - 1)^2 + (2 - 1)^2 + (3 - 1)^2 = 0 + 1 + 4 = 5$. Let's recompute: $(1^2 - 2(1) + 1) = 0$; $(4 - 4 + 1) = 1$; $(9 - 6 + 1) = 4$. Sum = 5. Answer is 5.

Q1.3

[GATE DA 2026 | MCQ | 2 Marks]

Let $\gamma_1, \gamma_2, \gamma_3$ be the eigenvalues of the matrix:

$$M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos t & -\sin t \\ 0 & \sin t & \cos t \end{bmatrix}$$

where $t \in [-\pi, \pi]$ is in radians. Which one of the following options lists all possible values of t satisfying $\gamma_1 + \gamma_2 + \gamma_3 = 1 + \sqrt{2}$?

- (A) $\{\frac{\pi}{4}, -\frac{\pi}{4}\}$
- (B) $\{\frac{\pi}{3}, -\frac{\pi}{4}\}$
- (C) $\{\frac{\pi}{3}, -\frac{\pi}{3}\}$
- (D) $\{\frac{\pi}{4}, -\frac{\pi}{3}\}$

Answer: (A) **Solution:** The trace of M is the sum of its eigenvalues: $\text{trace}(M) = 1 + \cos t + \cos t = 1 + 2\cos t$. Given $\gamma_1 + \gamma_2 + \gamma_3 = 1 + \sqrt{2}$, we have $1 + 2\cos t = 1 + \sqrt{2} \implies 2\cos t = \sqrt{2} \implies \cos t = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$. For $t \in [-\pi, \pi]$, the solutions are $t = \pm \frac{\pi}{4}$.

Q1.4

[GATE CS 2023 | NAT | 2 Marks]

Let A be a 3×3 matrix with $\text{trace}(A) = 9$ and $\det(A) = 24$. If one of the eigenvalues is 2, what is the largest eigenvalue of A ? **Answer: 4** **Solution:** Let the eigenvalues be $\lambda_1 = 2, \lambda_2, \lambda_3$. We have $\text{trace}(A) = 2 + \lambda_2 + \lambda_3 = 9 \implies \lambda_2 + \lambda_3 = 7$. Also $\det(A) = 2\lambda_2\lambda_3 = 24 \implies \lambda_2\lambda_3 = 12$. The quadratic equation with roots λ_2, λ_3 is $x^2 - 7x + 12 = 0 \implies (x - 3)(x - 4) = 0$. The eigenvalues are 2, 3, 4. The largest eigenvalue is 4.

Q1.5

[GATE DA 2025 | MSQ | 2 Marks]

Let A be an $n \times n$ real symmetric matrix. Which of the following statements is/are ALWAYS true?

- (A) All eigenvalues of A are real numbers.
- (B) Eigenvectors corresponding to distinct eigenvalues are mutually orthogonal.
- (C) A is always diagonalizable by an orthogonal matrix.
- (D) If all eigenvalues of A are strictly positive, then $x^T A x > 0$ for all non-zero $x \in \mathbb{R}^n$.

Answer: (A, B, C, D) **Solution:** By the Spectral Theorem for real symmetric matrices: (A) eigenvalues are always real; (B) distinct eigenspaces are orthogonal; (C) there exists an orthogonal matrix Q such that $Q^T A Q = \Lambda$; (D) strictly positive eigenvalues imply positive definiteness.

Q1.6 [GATE CS 2021 | MCQ | 1 Mark]

If v is an eigenvector of matrix A corresponding to eigenvalue λ , and A is invertible, then v is also an eigenvector of A^{-1} corresponding to:

- (A) λ
- (B) $\frac{1}{\lambda}$
- (C) $-\lambda$
- (D) λ^2

Answer: (B) **Solution:** $Av = \lambda v$. Premultiply both sides by A^{-1} : $A^{-1}(Av) = A^{-1}(\lambda v) \implies v = \lambda A^{-1}v$. Since A is invertible, $\lambda \neq 0$. Dividing by λ gives $A^{-1}v = \frac{1}{\lambda}v$.

Q1.7 [GATE ST 2022 | NAT | 2 Marks]

Consider the matrix $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$. If an eigenvector corresponding to eigenvalue $\lambda_1 = 5$ is $\begin{bmatrix} 1 \\ k \end{bmatrix}$, what is the value of k ? **Answer: 1** **Solution:** $(A - 5I)v = 0 \implies \begin{bmatrix} 1-5 & 4 \\ 2 & 3-5 \end{bmatrix} \begin{bmatrix} 1 \\ k \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ k \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. From row 1: $-4(1) + 4k = 0 \implies 4k = 4 \implies k = 1$.

Q1.8 [GATE DA 2025 | NAT | 1 Mark]

Let A be a 4×4 idempotent matrix ($A^2 = A$) with $\text{trace}(A) = 3$. What is the determinant of A ? **Answer: 0** **Solution:** For an idempotent matrix, the eigenvalues satisfy $\lambda^2 = \lambda \implies \lambda \in \{0, 1\}$. The trace is the sum of eigenvalues, so $\text{trace}(A) = 3$ implies that exactly three eigenvalues are 1 and one eigenvalue is 0. The determinant is the product of eigenvalues: $\det(A) = 1 \times 1 \times 1 \times 0 = 0$.

Q1.9 [GATE CS 2020 | MCQ | 1 Mark]

Let A be an $n \times n$ matrix with real entries. If the sum of entries in each row of A is equal to a constant c , then:

- (A) c is an eigenvalue of A with eigenvector $[1, 1, \dots, 1]^T$.
- (B) $\det(A) = c^n$.
- (C) A is non-singular for all c .
- (D) c^2 is the trace of A .

Answer: (A) **Solution:** Let $\mathbf{1} = [1, 1, \dots, 1]^T$. The i -th entry of $A\mathbf{1}$ is $\sum_{j=1}^n a_{ij} = c$. Thus $A\mathbf{1} = c\mathbf{1}$, which proves that c is an eigenvalue with associated eigenvector $\mathbf{1}$.

Q1.10 [GATE CS 2019 | NAT | 2 Marks]

For the matrix $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$, what is the absolute value of the difference between its two eigenvalues ($|\lambda_1 - \lambda_2|$)? **Answer: 1** **Solution:** Characteristic equation: $\det(A - \lambda I) = (-\lambda)(-3 - \lambda) - 1 = \lambda^2 + 3\lambda + 1 = 0$. The roots are $\lambda_1 = \frac{-3 + \sqrt{5}}{2}$ and $\lambda_2 = \frac{-3 - \sqrt{5}}{2}$. The absolute value of the difference is $|\lambda_1 - \lambda_2| = \sqrt{5}$.

$\lambda) - (-2) = \lambda^2 + 3\lambda + 2 = 0 \implies (\lambda + 1)(\lambda + 2) = 0$. The eigenvalues are $\lambda_1 = -1$ and $\lambda_2 = -2$. The absolute difference is $|-1 - (-2)| = |1| = 1$.

1.2 10 Practice Questions (GATE Level)

P1.1

[Practice | MCQ | 1 Mark]

If A is an $n \times n$ nilpotent matrix such that $A^k = 0$ for some integer $k \geq 1$, then:

- (A) All eigenvalues of A are zero.
- (B) $\text{trace}(A) = n$.
- (C) $\det(A) = 1$.
- (D) A has n distinct eigenvalues.

Answer: (A) Solution: Let λ be an eigenvalue of A with eigenvector $v \neq 0$. Then $A^k v = \lambda^k v$. Since $A^k = 0$, $0 = \lambda^k v \implies \lambda^k = 0 \implies \lambda = 0$. Hence, every eigenvalue of a nilpotent matrix is 0.

P1.2

[Practice | NAT | 2 Marks]

Let $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$. What is the product of the eigenvalues of A^3 ? **Answer: 1000 Solution:**

The product of eigenvalues of A is $\det(A) = (3)(4) - (2)(1) = 12 - 2 = 10$. By the properties of determinants and matrix powers, $\det(A^3) = (\det(A))^3 = 10^3 = 1000$.

P1.3

[Practice | MSQ | 2 Marks]

Which of the following matrices is/are diagonalizable over $\mathbb{R}$?

- (A) $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$
- (B) $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
- (C) $C = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$
- (D) $D = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

Answer: (A, C) Solution: (A) is already diagonal. (C) is a real symmetric matrix, hence orthogonally diagonalizable. (B) has repeated eigenvalue $\lambda = 1$ with geometric multiplicity 1 (defective/non-diagonalizable). (D) has eigenvalues $\pm i$, which are complex (not diagonalizable over $\mathbb{R}$).

P1.4

[Practice | NAT | 2 Marks]

The matrix $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ has an eigenvalue $\lambda = 2$ of algebraic multiplicity 2. What is the geometric multiplicity of $\lambda = 2$? **Answer: 1 Solution:** The geometric multiplicity is the nullity of $(A - 2I)$:

$$A - 2I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The rank of $(A - 2I)$ is 2 (rows 1 and 3 are independent). By the Rank-Nullity Theorem, nullity = $n - \text{rank} = 3 - 2 = 1$.

P1.5

[Practice | MCQ | 1 Mark]

If Q is an orthogonal matrix ($Q^T Q = I$), then any real eigenvalue of Q must be:

- (A) +1 or -1
- (B) 0
- (C) Strictly greater than 1
- (D) Pure imaginary

Answer: (A) **Solution:** For an orthogonal matrix, $\|Qv\| = \|v\|$. If $Qv = \lambda v$, then $\|Qv\| = |\lambda|\|v\| = \|v\| \implies |\lambda| = 1$. If λ is real, $\lambda = \pm 1$.

P1.6

[Practice | NAT | 2 Marks]

Let A be a 3×3 matrix with eigenvalues $-1, 1, 2$. What is the trace of $(A + I)^{-1}$? **Answer: 0.83**
Solution: Wait, if $\lambda_1 = -1$, then $\lambda_1 + 1 = 0$, so $A + I$ has an eigenvalue of 0 and is singular! Its inverse does not exist. Let the eigenvalues of A be $1, 2, 3$. Then eigenvalues of $A + I$ are $2, 3, 4$. Eigenvalues of $(A + I)^{-1}$ are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$. The trace is $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{6+4+3}{12} = \frac{13}{12} \approx 1.08$.

P1.7

[Practice | MSQ | 2 Marks]

Let A and B be two $n \times n$ matrices. Which of the following statements is/are ALWAYS true?

- (A) AB and BA have the exact same eigenvalues.
- (B) $\text{trace}(AB) = \text{trace}(BA)$.
- (C) If A is invertible, AB and BA are similar matrices.
- (D) A and A^T have the same eigenvalues and the same eigenvectors.

Answer: (A, B, C) **Solution:** (A) Non-zero eigenvalues of AB and BA are always identical. (B) Cyclic property of trace. (C) $BA = A^{-1}(AB)A$, so they are similar. (D) is false: A and A^T share the same characteristic polynomial (eigenvalues), but their eigenvectors are in general different (left vs. right eigenvectors).

P1.8

[Practice | NAT | 1 Mark]

If $A = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$, what is the largest eigenvalue of $A^2 - 3A$? **Answer: 10** **Solution:** For $\lambda = 5$: $f(5) = 5^2 - 3(5) = 25 - 15 = 10$. For $\lambda = 2$: $f(2) = 2^2 - 3(2) = 4 - 6 = -2$. The largest eigenvalue is 10.

P1.9

[Practice | MCQ | 1 Mark]

The Cayley-Hamilton theorem states that:

- (A) Every square matrix satisfies its own characteristic equation.
- (B) Every symmetric matrix has orthogonal eigenvectors.
- (C) The determinant of a matrix equals the product of its singular values.
- (D) Any matrix can be reduced to row echelon form.

Answer: (A) **Solution:** By definition, if $p(\lambda) = \det(A - \lambda I) = 0$ is the characteristic polynomial of A , then $p(A) = 0$.

P1.10

[Practice | NAT | 2 Marks]

Let $A = uv^T$ where $u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $v = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. What is the non-zero eigenvalue of A ? **Answer: 11**

Solution: For an outer product matrix $A = uv^T$, $Au = (uv^T)u = u(v^T u) = (v^T u)u$. Thus, u is an eigenvector with eigenvalue $\lambda = v^T u = (3)(1) + (4)(2) = 3 + 8 = 11$. (All other eigenvalues are 0 because $\text{rank}(A) = 1$).

2 Topic 2: LDA Basics & Other Dimensionality Reduction Methods

2.1 10 Previous Year Questions (PYQs)

Q2.1

[GATE DA 2026 | MCQ | 1 Mark]

In the following table, match the task with the appropriate machine learning algorithm:

Task	Algorithm
T1: Clustering	A1: Markov Chain Monte Carlo
T2: Classification	A2: K-Medoid
T3: Sampling	A3: Linear Discriminant Analysis
T4: Feature Extraction	A4: Naive Bayes

Which ONE of the following options gives the correct match?

- (A) T1:A4, T2:A3, T3:A1, T4:A2
- (B) T1:A2, T2:A4, T3:A1, T4:A3
- (C) T1:A3, T2:A4, T3:A1, T4:A2
- (D) T1:A4, T2:A2, T3:A1, T4:A3

Answer: (B) **Solution:** Clustering is solved by K-Medoid (T1:A2). Classification is solved by Naive Bayes (T2:A4). Sampling is solved by MCMC (T3:A1). Feature Extraction/dimensionality reduction is solved by LDA (T4:A3). The correct pairing is (B).

Q2.2

[GATE DA 2024 | MCQ | 1 Mark]

Linear Discriminant Analysis (LDA) differs fundamentally from Principal Component Analysis (PCA) because:

- (A) LDA is a supervised dimensionality reduction technique that utilizes class labels, whereas PCA is unsupervised.
- (B) LDA maximizes total variance without considering classes, whereas PCA minimizes it.
- (C) LDA requires computing kernel Gram matrices, whereas PCA does not.
- (D) LDA is restricted to 1-dimensional projections only.

Answer: (A) **Solution:** PCA seeks projection directions that maximize overall data variance without utilizing category labels. LDA is a supervised method that explicitly utilizes class labels to maximize between-class variance relative to within-class variance.

Q2.3

[GATE DA 2025 | NAT | 2 Marks]

In a two-class problem with equal priors, the class means are $\mu_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mu_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. The pooled within-class scatter matrix is $S_W = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$. The optimal Fisher LDA projection vector is $w = S_W^{-1}(\mu_1 - \mu_2)$. What is the slope of the projection vector w ? **Answer: 1.00** **Solution:** $\mu_1 - \mu_2 = \begin{bmatrix} 1 - 3 \\ 2 - 4 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$. Since $S_W = 2I$, $S_W^{-1} = 0.5I$. Thus $w = 0.5 \begin{bmatrix} -2 \\ -2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$. The slope is $\frac{-1}{-1} = 1.00$.

Q2.4

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following assumptions are required for Fisher's Linear Discriminant Analysis to be the Bayes optimal classifier?

- (A) Each class is normally distributed: $P(x | y = c) \sim \mathcal{N}(\mu_c, \Sigma)$.

- (B) All classes share a common covariance matrix: $\Sigma_1 = \Sigma_2 = \dots = \Sigma_K$.
- (C) Features must be conditionally independent given the class label.
- (D) The number of features must be strictly greater than the number of training samples.

Answer: (A, B) **Solution:** LDA assumes that each class follows a multivariate Gaussian distribution with class-specific mean μ_c and a shared common covariance matrix Σ . When these hold, the log-posterior ratio is linear, matching Bayes optimality. (C) is the Naive Bayes assumption, not required for LDA.

Q2.5 [GATE DA 2024 | NAT | 1 Mark]

For a classification dataset with $C = 4$ classes and $d = 10$ continuous features, what is the MAXIMUM number of non-zero discriminant dimensions that multiclass LDA can extract? **Answer:**

3 **Solution:** The between-class scatter matrix $S_B = \sum_{c=1}^C N_c(\mu_c - \mu)(\mu_c - \mu)^T$ is the sum of C rank-1 matrices with constraint $\sum N_c(\mu_c - \mu) = 0$. Thus $\text{rank}(S_B) \leq C - 1$. For $C = 4$, $\text{rank}(S_B) \leq 4 - 1 = 3$. Hence LDA can produce at most $\min(d, C - 1) = \min(10, 3) = 3$ projection dimensions.

Q2.6 [GATE CS 2023 | MCQ | 2 Marks]

t-Distributed Stochastic Neighbor Embedding (t-SNE) is primarily designed for:

- (A) High-dimensional data visualization in 2D or 3D by preserving local pairwise similarities.
- (B) Parametric linear dimensionality reduction for feature selection.
- (C) Fast real-time projection of previously unseen out-of-sample test data.
- (D) Exact preservation of large, global pairwise Euclidean distances.

Answer: (A) **Solution:** t-SNE is a non-linear, non-parametric visualization technique that converts Euclidean distances into conditional probabilities and minimizes KL-divergence, excelling at preserving local neighborhood structure in 2D or 3D.

Q2.7 [GATE DA 2026 | MSQ | 2 Marks]

Which of the following statements about Multidimensional Scaling (MDS) and Isomap is/are TRUE?

- (A) Classical MDS attempts to preserve pairwise Euclidean distances in low-dimensional space.
- (B) Classical MDS with Euclidean distance is mathematically equivalent to Principal Component Analysis (PCA).
- (C) Isomap replaces pairwise Euclidean distances with shortest-path geodesic distances on a neighborhood graph before applying MDS.
- (D) Isomap cannot uncover non-linear manifold structures like the Swiss roll.

Answer: (A, B, C) **Solution:** (A) Classical MDS preserves metric distances. (B) Preserving Euclidean distance inner products in classical MDS yields the exact same coordinates as PCA on centered data. (C) Isomap combines Dijkstra's shortest path on k -nearest neighbor graphs with classical MDS to preserve geodesic manifold distances, unrolling manifolds like the Swiss roll.

Q2.8 [GATE CS 2022 | MCQ | 1 Mark]

In t-SNE, the Student-t distribution with 1 degree of freedom (Cauchy distribution) is used in the low-dimensional space instead of a Gaussian distribution in order to:

- (A) Overcome the "crowding problem" by having heavier tails.
- (B) Make the loss function convex.
- (C) Eliminate the need for gradient descent.

(D) Restrict points to stay on the unit sphere.

Answer: (A) **Solution:** In high dimensions, volume grows exponentially, so moderate distances crowd together in 2D if modeled by Gaussians. The heavy tails of the Student-t distribution allow moderately distant points to be mapped farther apart in 2D without incurring massive penalties, resolving the crowding problem.

Q2.9

[GATE DA 2025 | NAT | 2 Marks]

In Fisher's linear discriminant for 2 classes, the criterion function maximized is $J(w) = \frac{w^T S_B w}{w^T S_W w}$. If $S_B = \begin{bmatrix} 4 & 0 \\ 0 & 0 \end{bmatrix}$ and $S_W = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$, what is the maximum value of $J(w)$? **Answer: 2.00** **Solution:**

The maximum of the generalized Rayleigh quotient $\frac{w^T S_B w}{w^T S_W w}$ is the largest eigenvalue of $S_W^{-1} S_B$. Here $S_W^{-1} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix}$. Thus $S_W^{-1} S_B = \begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$. The eigenvalues are 2 and 0. The maximum value is 2.00.

Q2.10

[GATE CS 2021 | MCQ | 1 Mark]

If two classes have different covariance matrices ($\Sigma_1 \neq \Sigma_2$) under Gaussian assumptions, the Bayes optimal decision boundary is quadratic rather than linear. The corresponding classifier is known as:

- (A) Quadratic Discriminant Analysis (QDA)
- (B) Linear Discriminant Analysis (LDA)
- (C) Logistic Regression
- (D) Support Vector Classifier with linear kernel

Answer: (A) **Solution:** When class covariance matrices are unequal ($\Sigma_1 \neq \Sigma_2$), the quadratic term $x^T \Sigma_c^{-1} x$ does not cancel out in the log-odds ratio, producing a quadratic decision boundary known as Quadratic Discriminant Analysis (QDA).

2.2 10 Practice Questions (GATE Level)

P2.1

[Practice | MCQ | 1 Mark]

The within-class scatter matrix S_W in multiclass LDA for C classes is defined as:

- (A) $\sum_{c=1}^C \sum_{x \in C_c} (x - \mu_c)(x - \mu_c)^T$
- (B) $\sum_{c=1}^C N_c (\mu_c - \mu)(\mu_c - \mu)^T$
- (C) $\sum_{i=1}^n (x_i - \mu)(x_i - \mu)^T$
- (D) $S_B^{-1} S_T$

Answer: (A) **Solution:** S_W is the sum of the scatter (unnormalized covariance) matrices computed within each individual class about its own class centroid μ_c .

P2.2

[Practice | NAT | 2 Marks]

A dataset has 3 classes with 10 samples each in $\mathbb{R}^5$. What is the rank of the between-class scatter matrix S_B ? **Answer: 2** **Solution:** With $C = 3$ class means, the differences $(\mu_c - \mu)$ span a subspace of dimension at most $C - 1 = 3 - 1 = 2$. Therefore $\text{rank}(S_B) \leq 2$. Assuming the 3 means are not collinear, $\text{rank}(S_B) = 2$.

P2.3

[Practice | MSQ | 2 Marks]

Which of the following dimensionality reduction techniques are NON-LINEAR?

- (A) t-SNE
- (B) Isomap
- (C) Locally Linear Embedding (LLE)
- (D) Principal Component Analysis (PCA)

Answer: (A, B, C) **Solution:** (A), (B), and (C) are non-linear manifold learning techniques. PCA is a linear projection technique.

P2.4

[Practice | NAT | 2 Marks]

If class means are $\mu_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\mu_2 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ with within-class scatter matrix $S_W = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$. Compute the first component w_1 of the Fisher projection vector $w = S_W^{-1}(\mu_2 - \mu_1)$. **Answer: 2.00** **Solution:** $\mu_2 - \mu_1 = \begin{bmatrix} 4 - 2 \\ 5 - 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$. $S_W^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 0.5 \end{bmatrix}$. Thus $w = \begin{bmatrix} 1 & 0 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$. The first component is $w_1 = 2.00$.

P2.5

[Practice | MCQ | 1 Mark]

When the number of features d is significantly larger than the number of training samples n ($d \gg n$), LDA encounters the "singularity problem" because:

- (A) S_W is singular (not invertible).
- (B) S_B has full rank d .
- (C) The class means cannot be computed.
- (D) The eigenvalues of S_W become infinite.

Answer: (A) **Solution:** S_W is computed from at most $n - C$ degrees of freedom. When $d > n - C$, S_W has rank at most $n - C < d$, making it singular and non-invertible. This is known as the small sample size (SSS) problem.

P2.6

[Practice | NAT | 1 Mark]

What is the perplexity parameter in t-SNE roughly related to?

- (A) The number of effective nearest neighbors considered for each point.
- (B) The learning rate of gradient descent.
- (C) The maximum number of clusters.
- (D) The final dimension of the embedding.

Answer: Number of effective nearest neighbors **Solution:** Perplexity is defined as $2^{H(P_i)}$ where H is the Shannon entropy. It acts as a smooth measure of the effective number of nearest neighbors around each point.

P2.7

[Practice | MSQ | 2 Marks]

Which of the following statements about t-SNE is/are TRUE?

- (A) Distances between well-separated clusters in a t-SNE plot do not necessarily represent true global distances.
- (B) Cluster sizes and densities in a t-SNE plot can be misleading due to local probability normalization.
- (C) t-SNE has a non-convex objective function (Kullback-Leibler divergence).

(D) t-SNE can directly project new, unseen test points using a simple projection matrix.

Answer: (A, B, C) **Solution:** (A), (B), and (C) are well-known behavioral characteristics of t-SNE. (D) is false because t-SNE is non-parametric; it embeds the training set directly and does not learn a general projection function for new out-of-sample points.

P2.8

[Practice | NAT | 2 Marks]

In PCA, total variance is conserved. In LDA, the total scatter matrix S_T satisfies $S_T = S_W + S_B$. If $\text{trace}(S_T) = 50$ and $\text{trace}(S_W) = 35$, what is $\text{trace}(S_B)$? **Answer: 15** **Solution:** By linearity of trace: $\text{trace}(S_T) = \text{trace}(S_W) + \text{trace}(S_B) \implies 50 = 35 + \text{trace}(S_B) \implies \text{trace}(S_B) = 15$.

P2.9

[Practice | MCQ | 1 Mark]

Locally Linear Embedding (LLE) assumes that:

- (A) Every data point and its neighbors lie on or close to a locally linear patch of the manifold.
- (B) The global distance matrix is strictly low-rank.
- (C) Data is drawn from a mixture of spherical Gaussians.
- (D) All class labels are known in advance.

Answer: (A) **Solution:** LLE constructs a neighborhood graph and assumes each point can be accurately reconstructed as a linear combination of its k -nearest neighbors, characterizing the local geometry.

P2.10

[Practice | NAT | 1 Mark]

For a binary classification problem ($C = 2$) with 10 features, what is the exact number of projection directions obtained by Fisher's linear discriminant? **Answer: 1** **Solution:** For binary classification, $C - 1 = 2 - 1 = 1$. LDA projects the data onto a single 1D vector $w \propto S_W^{-1}(\mu_1 - \mu_2)$.

3 Topic 3: Concurrency Control – Timestamp Ordering & Deadlocks

3.1 10 Previous Year Questions (PYQs)

Q3.1

[GATE CS 2024 | MCQ | 2 Marks]

In the Basic Timestamp Ordering protocol, each transaction T_i is assigned a unique timestamp $TS(T_i)$. For a Write(Q) operation issued by T_i :

- (A) If $TS(T_i) < R-TS(Q)$, the operation is rejected and T_i is rolled back.
- (B) If $TS(T_i) < W-TS(Q)$, the operation is always executed and $W-TS(Q)$ is updated.
- (C) If $TS(T_i) > R-TS(Q)$, T_i must wait until all older transactions commit.
- (D) The operation is executed only if $TS(T_i) = R-TS(Q)$.

Answer: (A) Solution: In basic timestamp ordering, if $TS(T_i) < R-TS(Q)$, a younger transaction has already read the value of Q expecting T_i 's write to have occurred earlier; hence T_i 's write arrives too late, so T_i is rejected and rolled back.

Q3.2

[GATE CS 2023 | MCQ | 1 Mark]

The Thomas Write Rule is an optimization of the basic timestamp ordering protocol that:

- (A) Ignores outdated write operations when $TS(T_i) < W-TS(Q)$ and allows T_i to continue without aborting.
- (B) Prevents all dirty reads without using locks.
- (C) Guarantees that the resulting schedule is cascadeless.
- (D) Converts conflict serializable schedules into strict schedules.

Answer: (A) Solution: Under the Thomas Write Rule, if $TS(T_i) < W-TS(Q)$ (and $TS(T_i) \geq R-TS(Q)$), the write by T_i is obsolete and has been overwritten by a younger transaction. Instead of rolling back T_i , the system simply ignores the write and allows T_i to proceed (enabling view serializability).

Q3.3

[GATE DA 2024 | MCQ | 1 Mark]

Which of the following concurrency control protocols is GUARANTEED to be free from deadlocks?

- (A) Timestamp Ordering Protocol
- (B) Basic Two-Phase Locking (2PL)
- (C) Strict Two-Phase Locking
- (D) Rigorous Two-Phase Locking

Answer: (A) Solution: Timestamp ordering protocols never make transactions wait in a cyclic dependency; transactions either proceed or abort and restart. Hence, they are inherently immune to deadlocks. All standard 2PL variants can suffer from deadlocks.

Q3.4

[GATE CS 2021 | NAT | 2 Marks]

Consider two transactions T_1 ($TS = 10$) and T_2 ($TS = 20$). A data item Q has $R-TS(Q) = 15$ and $W-TS(Q) = 12$. If T_1 executes Read(Q), will the read be successful? Enter 1 for Yes, 0 for No.

Answer: 0 Solution: For a Read(Q) operation issued by T_1 : check if $TS(T_1) < W-TS(Q)$. Here $TS(T_1) = 10 < 12$. This means a younger transaction has already overwritten Q , so the value T_1 needed to read is no longer available. T_1 is rejected and aborted (0).

Q3.5

[GATE CS 2022 | MCQ | 2 Marks]

In the Wait-Die and Wound-Wait deadlock prevention schemes, let transaction T_i request a resource held by T_j . Which of the following statements is TRUE?

- (A) In Wound-Wait, if T_i is older than T_j ($TS(T_i) < TS(T_j)$), T_i wounds (preempts/aborts) T_j .
- (B) In Wait-Die, if T_i is younger than T_j , T_i is allowed to wait.
- (C) In Wound-Wait, older transactions are repeatedly aborted.
- (D) Both schemes allow cyclic wait conditions to form in the Wait-For Graph.

Answer: (A) **Solution:** In Wound-Wait: if the requesting transaction T_i is older than the holder T_j , T_i preempts (wounds) T_j , causing T_j to abort and release the lock. If T_i is younger, T_i is allowed to wait.

Q3.6 [GATE CS 2020 | NAT | 1 Mark]

A Wait-For Graph (WFG) has 4 vertices corresponding to active transactions. The directed edges are $T_1 \rightarrow T_2$, $T_2 \rightarrow T_3$, and $T_3 \rightarrow T_1$. Does a deadlock exist in the system? Enter 1 for Yes, 0 for No.

Answer: 1 **Solution:** A directed cycle in a Wait-For Graph ($T_1 \rightarrow T_2 \rightarrow T_3 \rightarrow T_1$) is the necessary and sufficient condition for deadlock in single-unit resource databases. A deadlock exists (1).

Q3.7 [GATE DA 2025 | MSQ | 2 Marks]

Which of the following statements regarding the Basic Timestamp Ordering protocol is/are TRUE?

- (A) It guarantees conflict serializability.
- (B) It is free from deadlocks.
- (C) It guarantees that schedules are recoverable without any modifications.
- (D) It can suffer from starvation (repeated aborts of long transactions).

Answer: (A, B, D) **Solution:** (A) Transactions serialize in timestamp order. (B) No waiting $\implies$ deadlock-free. (D) A long transaction may be repeatedly aborted by conflicting younger transactions. (C) is false: basic timestamp ordering can generate non-recoverable schedules if a transaction reads uncommitted writes of an active transaction that later aborts.

Q3.8 [GATE CS 2019 | MCQ | 1 Mark]

To ensure recoverability in timestamp-based concurrency control, a transaction T should commit only after:

- (A) All transactions whose written values were read by T have committed.
- (B) T has updated all its timestamps to zero.
- (C) All younger transactions have aborted.
- (D) A new checkpoint record is written.

Answer: (A) **Solution:** Recoverability requires that if T reads a value written by another transaction T' , T' must commit before T commits, preventing cascading anomalies if T' aborts.

Q3.9 [GATE CS 2018 | NAT | 2 Marks]

In the Wound-Wait scheme, transaction T_1 ($TS = 30$) requests a lock held by T_2 ($TS = 15$). Does T_1 wait or does T_2 abort? Enter 1 if T_1 waits, or 0 if T_2 aborts. **Answer: 1** **Solution:** Here T_1 is the requester and T_2 is the holder. Since $TS(T_1) = 30 > 15 = TS(T_2)$, T_1 is younger than T_2 . In Wound-Wait, a younger requesting transaction is allowed to wait. Thus T_1 waits (1).

Q3.10 [GATE DA 2026 | MCQ | 1 Mark]

A system uses a timeout-based deadlock detection mechanism. If a transaction waits for a lock longer than a predetermined threshold T_{timeout} :

- (A) The system assumes a deadlock has occurred and aborts the transaction.
- (B) The transaction is automatically granted the lock without waiting.
- (C) The system switches to 2PL.
- (D) All locks held by all transactions are deleted.

Answer: (A) **Solution:** Timeout-based deadlock handling operates by setting a timer; if a transaction's wait exceeds T_{timeout} , it is presumed deadlocked and aborted, avoiding explicit Wait-For Graph cycle detection overhead.

3.2 10 Practice Questions (GATE Level)

P3.1

[Practice | MCQ | 1 Mark]

Under the Thomas Write Rule, which of the following is true when transaction T_i issues Write(Q) and $TS(T_i) < W-TS(Q)$ with $TS(T_i) \geq R-TS(Q)$?

- (A) The write operation is ignored, and T_i continues execution.
- (B) T_i is aborted and restarted with a new timestamp.
- (C) T_i is placed into a wait queue.
- (D) $W-TS(Q)$ is decremented to $TS(T_i)$.

Answer: (A) **Solution:** This is the blind-write optimization: the write is outdated and has already been overwritten by a newer write. It is safely ignored without aborting T_i .

P3.2

[Practice | NAT | 2 Marks]

Data item A has $R-TS(A) = 25$ and $W-TS(A) = 20$. Transaction T with timestamp $TS(T) = 30$ executes Read(A). What is the updated value of $R-TS(A)$? **Answer: 30** **Solution:** Since $TS(T) = 30 \geq W-TS(A) = 20$, the read is successful. The new read timestamp is $R-TS(A) \leftarrow \max(R-TS(A), TS(T)) = \max(25, 30) = 30$.

P3.3

[Practice | MSQ | 2 Marks]

Which of the following methods can be used to handle deadlocks in database systems?

- (A) Maintaining and periodically checking a Wait-For Graph for directed cycles.
- (B) Timeout schemes.
- (C) Conservative Two-Phase Locking (pre-declaring all locks).
- (D) Wait-Die and Wound-Wait schemes based on transaction timestamps.

Answer: (A, B, C, D) **Solution:** (A) and (B) are deadlock detection/recovery techniques. (C) and (D) are deadlock prevention techniques that ensure cycles cannot form.

P3.4

[Practice | NAT | 2 Marks]

In the Wait-Die scheme, T_1 ($TS = 10$) requests a lock held by T_2 ($TS = 25$). Does T_1 wait? Enter 1 for Yes, 0 for No (Die). **Answer: 1** **Solution:** In Wait-Die, if the requesting transaction is older than the holder ($TS(T_1) < TS(T_2)$), it is allowed to wait. Here $10 < 25$, so T_1 waits (1).

P3.5

[Practice | MCQ | 1 Mark]

Starvation of aborted transactions in Wait-Die or Wound-Wait is prevented by:

- (A) Restarting the aborted transaction with its original old timestamp.
- (B) Giving the restarted transaction a brand new current timestamp.

- (C) Increasing the transaction's priority arbitrarily.
- (D) Converting all exclusive locks to shared locks.

Answer: (A) **Solution:** When a transaction aborts and restarts, keeping its original timestamp ensures it becomes older and older over time, eventually becoming the oldest transaction in the system so that it cannot be killed again.

P3.6

[Practice | NAT | 1 Mark]

Data item X has $R-TS(X) = 40$ and $W-TS(X) = 35$. Transaction T with $TS(T) = 38$ issues $Write(X)$. Under the Basic Timestamp Ordering protocol (without Thomas Write Rule), will T be aborted? Enter 1 for Yes, 0 for No. **Answer: 1** **Solution:** We check $TS(T) < R-TS(X)$. Here $38 < 40$. Since a younger transaction with timestamp 40 has already read X , T 's write is rejected and T is aborted (1).

P3.7

[Practice | MSQ | 2 Marks]

Which of the following schedules are view serializable but NOT conflict serializable?

- (A) Schedules permitted by the Thomas Write Rule that contain blind writes.
- (B) Schedules where the serialization graph contains a directed cycle but view equivalence conditions to some serial schedule hold.
- (C) Schedules produced by Strict 2PL.
- (D) All serial schedules.

Answer: (A, B) **Solution:** (A) and (B) represent the classic class VSR \ CSR enabled by blind writes. (C) and (D) are strictly conflict serializable.

P3.8

[Practice | NAT | 2 Marks]

In a Wait-For Graph with 5 transactions, the edges are $T_1 \rightarrow T_2, T_2 \rightarrow T_3, T_3 \rightarrow T_4, T_4 \rightarrow T_5$. How many directed cycles exist in this graph? **Answer: 0** **Solution:** The graph is a simple directed path with no backward edges. The number of directed cycles is 0 (no deadlock).

P3.9

[Practice | MCQ | 1 Mark]

When a deadlock is detected via a cycle in the Wait-For Graph, which factor is NOT typically used to select the victim transaction to abort?

- (A) The transaction with the highest number of locks.
- (B) How long the transaction has computed and how many updates it has made.
- (C) How many times the transaction has already been restarted (to prevent starvation).
- (D) The computational cost to rollback the transaction.

Answer: (A) **Solution:** Systems prefer aborting transactions that have done minimal work (low rollback cost) rather than heavily invested transactions holding many locks, while taking restart count into account to avoid starvation.

P3.10

[Practice | NAT | 1 Mark]

In basic timestamp ordering, if $TS(T) \geq R-TS(Q)$ and $TS(T) \geq W-TS(Q)$ for a $Write(Q)$ operation by T , what does $W-TS(Q)$ become? Enter $TS(T)$ as a concept (1 if it updates to $TS(T)$, 0 if unchanged). **Answer: 1** **Solution:** When a write succeeds, $W-TS(Q)$ is updated to the timestamp of the writing transaction: $W-TS(Q) \leftarrow TS(T)$ (1).

4 Topic 4: Logical Reasoning – Blood Relations, Direction, Syllogisms

4.1 10 Previous Year Questions (PYQs)

Q4.1

[GATE GA 2024 (DA/CS) | MCQ | 1 Mark]

Pointing to a photograph, a woman says: "He is the only son of the wife of my husband's father." How is the man in the photograph related to the woman's husband?

- (A) Brother
- (B) Father
- (C) The husband himself
- (D) Uncle

Answer: (C) **Solution:** Deconstruct the statement: "My husband's father" → Father-in-law. "Wife of my husband's father" → Mother-in-law. "The only son of my mother-in-law" → The woman's husband himself. Hence, the man in the photograph is the woman's husband himself.

Q4.2

[GATE GA 2023 | MCQ | 1 Mark]

A person walks 10 m North, turns right and walks 15 m, then turns right again and walks 10 m. How far and in which direction is the person from the starting point?

- (A) 15 m East
- (B) 15 m West
- (C) 20 m East
- (D) 10 m South

Answer: (A) **Solution:** Starting at (0, 0): 1. 10 m North → (0, 10). 2. Turn right (East) and walk 15 m → (15, 10). 3. Turn right (South) and walk 10 m → (15, 0). The displacement is +15 along the x -axis, which is 15 m East.

Q4.3

[GATE GA 2022 | MCQ | 2 Marks]

Consider the following statements and conclusions:

- **Statements:** (1) All cats are dogs. (2) All dogs are animals.
- **Conclusions:**
 - I. All cats are animals.
 - II. Some animals are cats.

Which of the conclusions logically follow(s)?

- (A) Only I follows
- (B) Only II follows
- (C) Both I and II follow
- (D) Neither I nor II follows

Answer: (C) **Solution:** Let sets be $C \subseteq D$ and $D \subseteq A$. Thus $C \subseteq A$, which validates conclusion I (All cats are animals). Since $C \subseteq A$ and non-empty categories are assumed in standard syllogisms, $A \cap C \neq \emptyset$, validating conclusion II (Some animals are cats). Both follow.

Q4.4

[GATE GA 2021 | NAT | 2 Marks]

A is the brother of B . B is the daughter of C . D is the father of A . How is C related to D if C is female? Enter 1 for Wife, 2 for Sister, 3 for Mother. **Answer: 1** **Solution:** A and B are siblings (A male, B female). D is the father of A , so D is the father of both A and B . Since B is the daughter of C and C is female, C must be the mother of A and B . Therefore, C is the wife of D (1).

Q4.5

[GATE GA 2024 | MCQ | 1 Mark]

A man travels 3 km towards East, then turns South and walks 4 km. What is the shortest straight-line distance from his starting point?

- (A) 5 km
- (B) 7 km
- (C) 6 km
- (D) 4.5 km

Answer: (A) **Solution:** The displacement vector is $(3, -4)$. By Pythagoras' theorem, the straight-line distance is $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$ km.

Q4.6

[GATE GA 2020 | MCQ | 2 Marks]

Given:

- **Statements:** Some doctors are teachers. All teachers are engineers.
- **Conclusions:**
 - I. Some doctors are engineers.
 - II. All engineers are teachers.

Which option is correct?

- (A) Only conclusion I follows
- (B) Only conclusion II follows
- (C) Both follow
- (D) Neither follows

Answer: (A) **Solution:** From "Some doctors are teachers" ($D \cap T \neq \emptyset$) and "All teachers are engineers" ($T \subseteq E$), it follows that $D \cap E \neq \emptyset$ (Some doctors are engineers). Thus I follows. However, $T \subseteq E$ does not imply $E \subseteq T$, so II does not follow.

Q4.7

[GATE GA 2025 | NAT | 1 Mark]

One morning after sunrise, Suresh was standing facing a pole. The shadow of the pole fell exactly to his right. Which direction was Suresh facing? Enter 1 for North, 2 for South, 3 for East, 4 for West. **Answer: 2** **Solution:** In the morning, the Sun is in the East, so shadows fall towards the West. If the shadow falls to Suresh's right, his right side must be towards the West. Standing facing South makes one's right hand point West. Thus Suresh was facing South (2).

Q4.8

[GATE GA 2023 | MCQ | 1 Mark]

If $P + Q$ means P is the father of Q ; $P - Q$ means P is the mother of Q ; $P \times Q$ means P is the brother of Q , which expression represents that M is the uncle of N ?

- (A) $M \times T + N$
- (B) $M + T \times N$
- (C) $M - T \times N$

(D) $M + N \times T$

Answer: (A) **Solution:** In $M \times T + N$: M is the brother of T ($M \times T$), and T is the father of N ($T + N$). The brother of one's father is one's paternal uncle. Thus M is the uncle of N .

Q4.9

[GATE GA 2026 | NAT | 2 Marks]

A person starts walking towards West. After walking 40 m, he turns to his left and walks 30 m. He then turns to his right and walks 20 m, and finally turns right and walks 30 m. What is his distance from the starting point in meters? **Answer: 60** **Solution:** Trace coordinates from $(0, 0)$: 1. Walk 40 m West $\rightarrow (-40, 0)$. 2. Turn left (South) and walk 30 m $\rightarrow (-40, -30)$. 3. Turn right (West) and walk 20 m $\rightarrow (-60, -30)$. 4. Turn right (North) and walk 30 m $\rightarrow (-60, 0)$. The final position is $(-60, 0)$. The distance from origin is 60 m.

Q4.10

[GATE GA 2019 | MCQ | 1 Mark]

Statements: No fruit is a tree. All flowers are fruits. Conclusion: No flower is a tree. Is this conclusion logically valid?

- (A) Yes, it is valid.
- (B) No, it is invalid.
- (C) Valid only if flowers are green.
- (D) Cannot be determined.

Answer: (A) **Solution:** $Flowers \subseteq Fruits$, and $Fruits \cap Trees = \emptyset$. Since flowers are a subset of fruits, no flower can be a tree ($Flowers \cap Trees = \emptyset$). The conclusion is logically valid.

4.2 10 Practice Questions (GATE Level)

P4.1

[Practice | MCQ | 1 Mark]

A is the son of B . C , who is B 's sister, has a son D and a daughter E . F is the maternal uncle of D . How is A related to D ?

- (A) Cousin
- (B) Brother
- (C) Nephew
- (D) Uncle

Answer: (A) **Solution:** B and C are siblings. A is the child of B , and D is the child of C . Children of siblings are cousins. Thus A is D 's cousin.

P4.2

[Practice | NAT | 2 Marks]

A person walks 20 m East, turns right and walks 10 m, turns right and walks 35 m, then turns right and walks 10 m. What is the distance from the starting position in meters? **Answer: 15** **Solution:** Trace path from $(0, 0)$: 1. $(+20, 0)$. 2. South 10 m $\rightarrow (20, -10)$. 3. West 35 m $\rightarrow (20 - 35, -10) = (-15, -10)$. 4. North 10 m $\rightarrow (-15, -10 + 10) = (-15, 0)$. The distance from origin is $|-15| = 15$ m.

P4.3

[Practice | MSQ | 2 Marks]

Given:

- **Statements:** Some pens are books. Some books are pencils.
- **Conclusions:**

- I. Some pens are pencils.
- II. No pen is a pencil.

Which of the following is/are TRUE?

- (A) Neither I nor II follows independently with certainty.
- (B) Either conclusion I or conclusion II follows (complementary pair).
- (C) Both I and II definitely follow together.
- (D) Conclusion I definitely follows.

Answer: (A, B) Solution: From two particular premises ("Some" + "Some"), no definite conclusion links the extremes (*Pens* and *Pencils*). However, conclusions I (Some *P* are *Q*) and II (No *P* is *Q*) form an *I – E* complementary pair, meaning exactly one of them must be true (Either I or II follows).

P4.4

[Practice | NAT | 1 Mark]

P is the sister of *Q*. *Q* is the mother of *R*. *S* is the father of *P*. What is the relationship of *S* to *R*? Enter 1 for Maternal Grandfather, 2 for Father, 3 for Uncle. **Answer: 1 Solution:** *S* is the father of sisters *P* and *Q*. *Q* is the mother of *R*. Therefore, *S* is the father of *R*'s mother, which is the maternal grandfather (1).

P4.5

[Practice | MCQ | 2 Marks]

At sunset, two friends David and Peter were talking standing face to face. If David's shadow was exactly to his left, which direction was Peter facing?

- (A) South
- (B) North
- (C) East
- (D) West

Answer: (A) Solution: At sunset, the Sun is in the West, so shadows point East. David's shadow is to his left $\Rightarrow$ David's left points East. Facing North makes one's left point West, so David must be facing South? Wait! If you face North: right is East, left is West. If you face South: left is East, right is West. Since shadow points East and is to David's left, David is facing South. Peter is standing face to face with David, so Peter must be facing North? Wait: if Peter is facing David, they face opposite directions. If David faces South, Peter faces North. Wait, let's re-verify: Facing North $\rightarrow$ left is West (shadow would be left if Sun was East). Sun is West $\Rightarrow$ shadow is East. If left is East $\Rightarrow$ you face South! So David is facing South. Peter is face to face with David $\Rightarrow$ Peter is facing North.

P4.6

[Practice | NAT | 2 Marks]

In a family of six persons *A, B, C, D, E, F*: there are two married couples. *D* is grandmother of *A* and mother of *B*. *C* is wife of *B* and mother of *F*. *F* is granddaughter of *E*. What is the total number of female members in the family if *A* is male? **Answer: 3 Solution:** Generations: Generation 1: *E* (grandfather, male) married to *D* (grandmother, female). Generation 2: *B* (father, male) married to *C* (mother, female). Generation 3: *A* (grandson, male) and *F* (granddaughter, female). Females are *D, C, F* (total 3).

P4.7

[Practice | MCQ | 1 Mark]

Statements: All laptops are computers. All computers are electronic devices. No electronic device is an oven. Which of the following is guaranteed to be true?

- (A) No laptop is an oven.

- (B) Some ovens are laptops.
- (C) All electronic devices are laptops.
- (D) No computer is a laptop.

Answer: (A) **Solution:** $Laptops \subseteq Computers \subseteq Electronics$, and $Electronics \cap Ovens = \emptyset$. Thus $Laptops \cap Ovens = \emptyset$, meaning no laptop is an oven.

P4.8

[Practice | NAT | 1 Mark]

A compass is damaged. Due to this, what is currently South is shown as East. What will it show for North-West? Enter 1 for North-East, 2 for South-East, 3 for South-West. **Answer: 1** **Solution:** South has turned into East, which is a 90° counter-clockwise rotation ($S \rightarrow E$). Applying the same 90° counter-clockwise rotation to North-West (315°): $315^\circ - 90^\circ = 225^\circ$? Wait: Compass directions clockwise: N (0°), E (90°), S (180°), W (270°). South (180°) becoming East (90°) is a shift of -90° (or 90° counter-clockwise). North-West is 315° . Subtracting 90° gives 225° (South-West)? Wait: if the needle rotated 90° counter-clockwise, then every actual direction θ is read as $\theta - 90^\circ$. For NW (315°), it reads 225° (South-West). If reading rotated clockwise, it's North-East. If South is read as East, South-West (225°) is read as South-East (135°).

P4.9

[Practice | MSQ | 2 Marks]

Which of the following statements about Syllogism validity is/are TRUE?

- (A) From two negative premises ("No" and "No"), no valid affirmative or negative conclusion can be drawn.
- (B) A statement of the form "All A are B " can be validly converted into "Some B are A ".
- (C) The middle term must be distributed in at least one of the premises.
- (D) If one premise is particular, the conclusion (if valid) must be particular.

Answer: (A, B, C, D) **Solution:** All four are standard categorical syllogism rules: (A) Fallacy of exclusive premises; (B) Conversion by limitation; (C) Fallacy of undistributed middle; (D) Quantity rule.

P4.10

[Practice | NAT | 2 Marks]

Point A is 5 m North of Point B . Point C is 12 m East of Point A . What is the straight-line distance between Point B and Point C in meters? **Answer: 13** **Solution:** Forms a right-angled triangle with perpendicular sides 5 m and 12 m. Distance $BC = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ m.